

ARCO 17 SW, ID, USA

WMO: 725790

Lat: 43.4621N

Lon: 113.5560W

Elev: 5920

StdP: 11.81

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 04126

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.9	3.3	-7.5	4.6	1.0	-3.5	5.7	6.2	25.4	29.6	22.1	29.3	2.4	N/A	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.5	89.1	55.3	86.6	54.5	83.9	53.9	58.1	79.1	56.9	78.7	55.7	77.9	9.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
51.7	71.2	61.1	49.1	64.4	61.0	46.7	58.8	60.9	27.7	79.3	26.8	78.5	26.0	77.7	63.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.6	18.0	15.8	DB	-6.8	93.9	4.3	2.0	-9.8	95.3	-12.3	96.5	-14.7	97.6	-17.8	99.0	(4)
			WB	-6.5	60.6	5.2	1.4	-10.3	61.6	-13.3	62.4	-16.3	63.2	-20.1	64.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.5	19.3	22.4	32.1	41.7	50.6	60.1	71.0	68.4	58.6	44.6	31.3	20.4
	DBStd	19.50	8.33	8.53	8.43	7.60	8.02	8.17	5.31	5.86	8.48	8.90	9.15	8.86
	HDD50	4379	951	774	556	269	94	15	0	0	27	213	561	919
	HDD65	8263	1416	1194	1020	700	449	186	15	35	219	634	1011	1384
	CDD50	1999	0	0	1	19	112	317	650	572	283	45	0	0
	CDD65	409	0	0	0	0	2	39	200	142	25	1	0	0
	CDH74	4754	0	0	0	2	43	513	2292	1575	321	8	0	0
	CDH80	1555	0	0	0	0	2	137	861	502	52	1	0	0
Wind	WSAvg	5.9	4.5	5.9	6.6	6.9	6.7	6.8	6.3	6.0	5.9	5.6	5.3	4.7
Precipitation	PrecAvg	11.1	1.3	0.9	0.8	1.1	1.4	1.2	0.5	0.6	0.6	0.8	0.7	1.3
	PrecMax	16.5	2.4	2.8	2.8	2.0	3.3	3.4	1.1	1.2	1.2	2.1	2.0	3.4
	PrecMin	7.5	0.1	0.0	0.0	0.3	0.1	0.1	0.0	0.1	0.1	0.1	0.1	0.3
	PrecStd	2.8	0.7	0.7	0.7	0.4	1.0	0.9	0.4	0.4	0.3	0.6	0.5	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.2	46.6	60.3	70.4	78.7	89.7	92.9	91.0	85.6	73.4	57.2	42.2
		MCWB	33.6	38.1	42.3	47.4	50.8	57.0	57.2	55.2	53.9	51.1	43.6	35.8
	2%	DB	35.5	40.9	54.3	65.1	74.4	83.7	90.2	87.9	80.9	67.7	52.5	38.5
		MCWB	31.5	34.5	40.3	44.9	49.8	54.3	55.7	54.4	52.3	47.8	40.9	35.2
	5%	DB	32.7	37.4	49.0	60.4	70.3	80.0	87.7	85.6	77.8	63.0	47.9	35.4
		MCWB	29.9	32.4	37.8	42.9	48.8	52.8	55.0	53.9	51.2	45.6	38.6	32.5
	10%	DB	30.4	34.6	44.3	55.9	66.2	76.1	85.3	82.8	74.4	58.9	44.0	32.7
		MCWB	28.1	30.9	36.0	41.2	47.4	51.6	54.4	53.3	50.1	44.2	36.5	30.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.5	39.1	43.9	49.0	53.4	59.3	59.9	59.2	59.2	51.4	44.8	37.8
		MCDB	38.1	45.9	57.3	68.6	68.3	85.7	82.0	74.8	70.9	67.6	57.0	40.4
	2%	WB	33.1	36.8	41.2	46.3	51.3	56.8	58.1	57.4	55.8	49.9	41.9	35.6
		MCDB	35.8	42.2	51.6	62.0	67.9	80.9	81.4	75.7	71.6	62.8	51.5	38.0
	5%	WB	31.3	34.4	39.0	44.0	49.7	54.7	57.1	56.2	53.8	47.8	39.6	33.0
		MCDB	33.1	38.3	47.9	58.6	66.0	76.6	80.9	76.5	71.5	60.1	46.9	35.4
	10%	WB	29.1	32.3	37.2	41.9	48.2	53.0	56.1	55.0	51.9	45.5	37.4	30.5
		MCDB	31.1	35.7	43.8	53.9	63.3	73.7	79.8	76.9	69.8	56.6	43.6	32.4
Mean Daily Temperature Range	5% DB	MDBR	11.9	12.2	14.5	19.4	21.0	23.5	25.5	25.2	23.2	19.4	15.3	11.9
		MCDBR	13.8	14.2	20.9	25.6	25.6	27.7	27.9	27.7	26.6	23.5	19.3	13.0
		MCWBR	11.5	10.4	11.6	12.8	11.2	10.5	10.2	10.4	10.7	11.1	11.0	10.9
	5% WB	MCDBR	13.1	14.2	18.1	23.2	23.1	26.2	25.7	24.7	22.4	20.6	18.8	12.5
		MCWBR	11.2	10.2	10.4	12.3	10.5	9.9	9.6	9.5	9.3	10.6	11.1	10.6
Clear-Sky Solar Irradiance	taub	0.229	0.238	0.255	0.283	0.296	0.298	0.307	0.307	0.286	0.264	0.245	0.230	
	taud	2.453	2.489	2.574	2.521	2.532	2.557	2.547	2.520	2.590	2.630	2.611	2.516	
	Ebn at Noon	297	313	318	313	309	308	304	301	302	299	291	288	
	Edh at Noon	22	26	27	31	32	31	31	31	27	23	19	19	
All-Sky Solar Radiation	RadAvg	555	849	1276	1703	2029	2367	2390	2065	1628	1099	676	458	
	RadStd	63	112	118	120	192	178	125	118	108	74	52	51	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page